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CICLO XXXIX

TIME SERIES ANALYSIS FOR SEISMIC WAVE DETECTION AND PREDICTION

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Chapter 1

Introduction

Earth's litosphere is a very complex and dynamic system as it is constantly subject to stresses and deformations in response to the forces and rheological properties of the materials acting on it. The lithosphere is composed of the crust and the uppermost part of the mantle and it is divided into tectonic plates that float on the semi-fluid asthenosphere. The relative motion of these plates is responsible for the formation of mountain ranges, ocean basins, earthquakes and volcanic activity. The study of the lithosphere and its dynamics is of great importance as it helps us to understand the processes that shape the Earth's surface and to predict natural hazards such as earthquakes and volcanic eruptions.

After the earthquake in L'Aquila, Italy, in 2009, which caused 309 deaths and left 65,000 people homeless, the Italian government decided to establish the Istituto Nazionale di Geofisica e Vulcanologia (INGV) to monitor and study seismic activity in the country. The INGV has deployed a network of over 400 seismometers throughout Italy to detect and locate earthquakes and to study the propagation of seismic waves through the Earth's crust. These vast amounts of data collected by the seismometers are processed and analyzed by seismologists to estimate the seismic hazard of the region. Traditional methods, relying on physics-based models and empirical relationships, have significantly advanced our understanding of earthquake processes. However, these methods have some limitations, such as the need for simplifying assumptions and the inability to capture the complexity of the lithosphere. Furthermore, the traditional methods of detecting and locating earthquakes rely on the manual inspection of seismic data by seismologists, which is a time-consuming and subjective process. The inherent complexity, nonlinearity, and variability of seismic phenomena

have motivated the exploration of new methodologies that can handle large amounts of data and uncover patterns that are difficult to detect using conventional techniques.

In recent years, there has been an increasing interest in the use of machine learning (ML) techniques to analyze seismic data and to improve the detection and location of earthquakes. Machine learning algorithms have been shown to be effective in detecting seismic events, in classifying seismic signals and in estimating the magnitude and location of earthquakes. The ability of ML models to recognize patterns, make inferences, and generalize from data makes them well-suited for the complex and data-rich field of seismology. Seismological datasets are often vast and multidimensional, offering a fertile ground for the application of ML techniques.

In this thesis, we explore the use of ML techniques for the detection and prediction of seismic waves. We focus on the analysis of seismic waveforms recorded by seismometers in the North Eastern Italy region known as AdriaArray. Our goal is to develop ML models that can automatically detect earthquakes from seismic waveforms and predict the arrival times of seismic waves in real-time. This research, as implied by the Piano Nazionale di Ripresa e Resilienza (PNRR) objectives, has the potential to improve the efficiency and accuracy of the seismic monitoring system in Italy and to enhance the understanding of seismic phenomena through digital innovation. Furthermore, we develop a cloud-based platform publicly available, infrastructure provided by the Istituto Nazionale di Oceanografia e di Geofisica Sperimetale (OGS), for the scientific community to access and analyze seismic data, to train ML models, and to share the results of the analysis under the Terabit Network for Research and Academic Big Data in Italy (TeRABIT) project.

Chapter 2

Literature Review

2.1 PhaseNet

PhaseNet is a deep learning model for the detection of seismic phases in continuous waveforms. The model is based on a convolutional neural network (CNN) architecture that takes as input a raw waveform and outputs the probability of the presence of a seismic phase at each time step. The model is trained on a large dataset of seismic waveforms with labeled phase picks and is able to learn the complex patterns in the data that are indicative of seismic phases. The model is designed to be robust to noise and can generalize well to unseen data. PhaseNet has been shown to outperform traditional methods for phase detection and is able to detect seismic phases with high accuracy and low latency.

2.2 EQTransformer

EQTransformer is a deep learning model for the detection and classification of seismic signals in continuous waveforms. The model is based on a convolutional neural network (CNN) architecture that takes as input a raw waveform and outputs the probability of the presence of a seismic signal at each time step. The model is trained on a large dataset of seismic waveforms with labeled signal picks and is able to learn the complex patterns in the data that are indicative of seismic signals. The model is designed to be robust to noise and can generalize well to unseen data. EQTransformer has been shown to outperform traditional methods for signal detection and classification and is able to detect seismic signals with high accuracy and low latency.

2.3 GaMMA

GaMMA is a probabilistic model for the detection and classification of seismic signals in continuous waveforms. The model is based on a Gaussian mixture model (GMM) architecture that takes as input a set of features extracted from the waveform and outputs the probability of the presence of a seismic signal. The model is trained on a large dataset of seismic waveforms with labeled signal picks and is able to learn the complex patterns in the data that are indicative of seismic signals. The model is designed to be robust to noise and can generalize well to unseen data. GaMMA has been shown to outperform traditional methods for signal detection and classification and is able to detect seismic signals with high accuracy and low latency.

Chapter 3

Data Description

	Pred. P	**Pred. void**
True P	TP	FN
True void	FP	TN

	Pred. S	**Pred. void**
True S	TP	FN
True void	FP	TN

Table 3.1: Confusion matrix for the binary classification of P and S waves, left and right, respectively.

	Pred. P	**Pred. S**	**Pred. void**
True P	cTP	pTP	FN
True S	pTP	cTP	FN
True void	FP	FP	TN

Table 3.2: Confusion matrix for the classification of seismic waveforms.

Chapter 4

Time Series Analysis

Chapter 5

Seismic Wave Detection

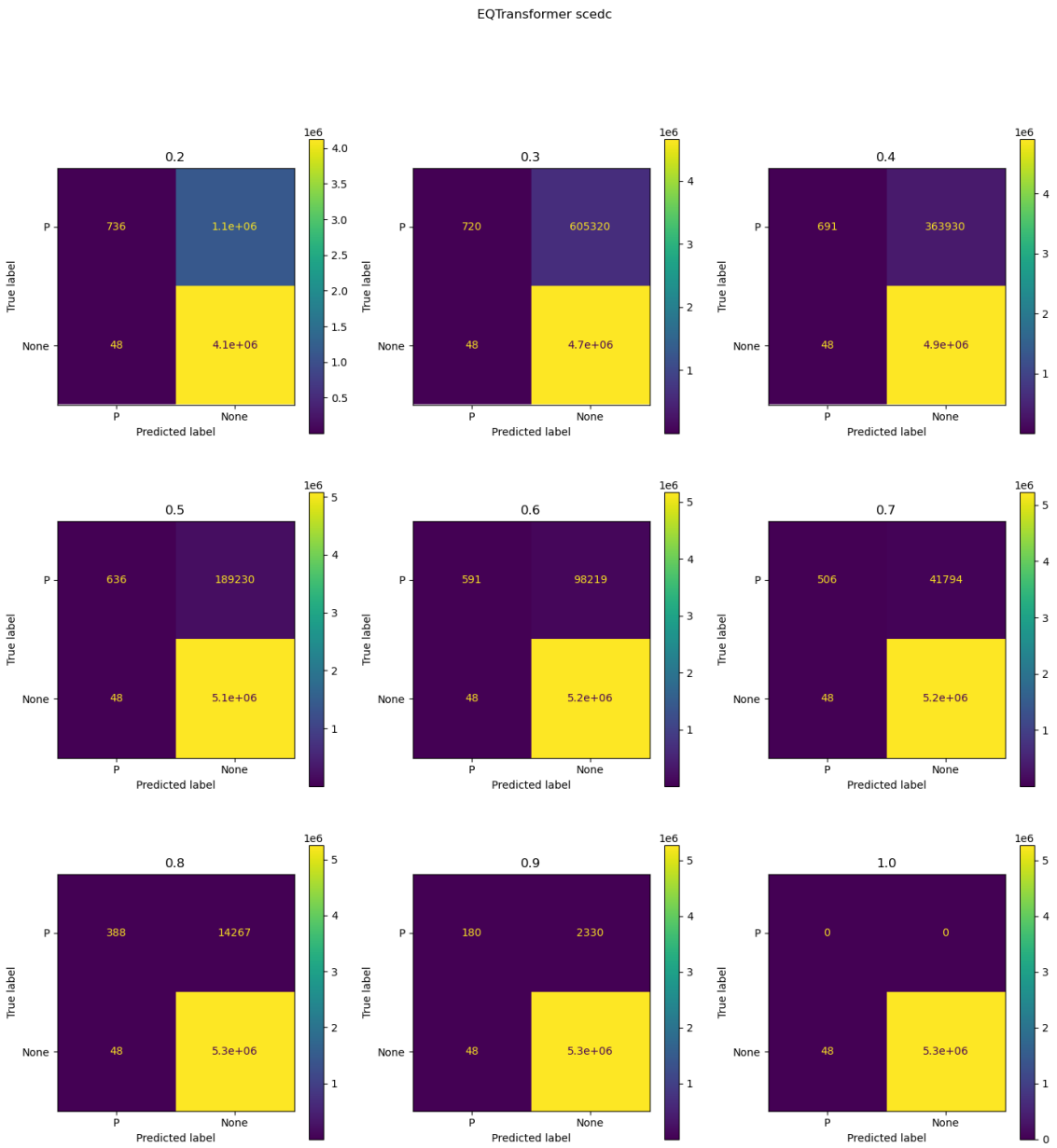

Figure 5.1: Result of the EQTransformer architecture trained on the Southern California Earthquake Data Center (SCEDC) dataset.

Chapter 6

Seismic Wave Prediction

Chapter 7

Conclusions